24 Oct 2026 21:34:48 - optim_solvers.sm
Created using a free version of SMath Studio

OptimWrapper - complete solver reference (CAS-defined problems)

2026-10-24

OptimWrapper - every solver, user-defined problems

GUID 3c9f7e0a-c2b1-4eae-ab68-9ea2c518c510, v1.0.0.0 (OptimLib 3.1.3 bridge).

The plugin ships NO built-in test problems: every objective / constraint / system

below is defined here in SMath CAS. Solver result layout: [x1; ...; xn; f; iterations; ok]

f = el(result, n+1); iterations = el(result, n+2); ok = el(result, n+3) (1 = converged).

All values are unitless. DE/PSO take explicit box bounds (p_min, p_max).

Run Calculate All (Ctrl+Shift+F9) if values are not filled in on open.

1. The 13 optimisation algorithms

LOCAL DETERMINISTIC - unconstrained scalar objective f(x):

0 bfgs Quasi-Newton (BFGS inverse-Hessian update + line search); robust default.

1 lbfgs Limited-memory BFGS: low memory, good for many variables.

2 cg Nonlinear conjugate gradient: little memory; needs a good line search.

3 gd Gradient descent (steepest descent): simple; slow/diverges with a fixed step.

4 newton Newton with a numerical Hessian: fast near a minimum, O(n^2) memory.

5 nm Nelder-Mead simplex: derivative-free; robust but slow in high dimension.

GLOBAL STOCHASTIC - population 200, require bounds:

6 de Differential evolution: mutation/crossover/selection global search.

7 de_prmm DE with pro-active mutation: extra diversity to avoid premature convergence.

8 pso Particle swarm: particles follow personal/global bests.

9 pso_dv PSO with differential variation: more explorative, noisier.

CONSTRAINED:

10 sumt Sequential Unconstrained Minimisation (penalty) for ci(x) <= 0.

ROOT FINDING - square system F(x) = 0:

11 broyden Broyden quasi-Newton root finder (secant update of the Jacobian).

12 broyden_df Broyden dual-form variant (same goal, different update).

Not for commercial use

Created using a free version of SMath Studio

2. Problems defined in SMath CAS (objectives, a constraint system, two root systems)

Rosenbrock objective; minimum (1, 1), f = 0.

$$f_rosen(u, v) := 1 - u^2 + 100 \cdot v - u^2^2$$

Booth objective; minimum (1, 3), f = 0.

$$f_booth(u, v) := u + 2 \cdot v - 7^2 + 2 \cdot u + v - 5^2$$

Convex quadratic; minimum (2.25, -4.75), f = 0.

$$f_quad(u, v) := u - 2.25^2 + v + 4.75^2$$

4-D sphere; minimum (0, 0, 0, 0), f = 0.

$$f_sph4(u, v, w, z) := u^2 + v^2 + w^2 + z^2$$

Rastrigin multimodal objective; global minimum (0, 0), f = 0.

$$f_rast(u, v) := 20 + u^2 - 10 \cdot \cos(2 \cdot \pi \cdot u) + v^2 - 10 \cdot \cos(2 \cdot \pi \cdot v)$$

Constrained [f; c1] with c1 = -2u - v + 14 <= 0; minimum (5, 4).

$$c_quad(u, v) := \begin{bmatrix} u - 5^2 + v - 4^2 \\ 0 - 2 \cdot u - v + 14 \end{bmatrix}$$

Root system u^2+v^2=1, u=v -> (0.7071068, 0.7071068).

$$sys1(u, v) := \begin{bmatrix} u^2 + v^2 - 1 \\ u - v \end{bmatrix}$$

Root system u*exp(u)=1 (omega constant), u=v -> (0.5671433, 0.5671433).

$$sys2(u, v) := \begin{bmatrix} u \cdot \exp(u) - 1 \\ u - v \end{bmatrix}$$

Rosenbrock start [-1.2; 1].

$$s_rn := \begin{bmatrix} 0 - 1.2 \\ 1 \end{bmatrix}$$

Zero start [0; 0].

$$s_r0 := \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Booth start [0; 0].

$$s_b := \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Rastrigin start [3; 3].

$$s_ras := \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

Not for commercial use

Created using a free version of SMath Studio

Rastrigin lower bounds [-5.12; -5.12].

$$lo_r := \begin{bmatrix} 0 - 5.12 \\ 0 - 5.12 \end{bmatrix}$$

Rastrigin upper bounds [5.12; 5.12].

$$hi_r := \begin{bmatrix} 5.12 \\ 5.12 \end{bmatrix}$$

Constrained start [0; 0].

$$s_c := \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Circle-line start [0.7; 0.7].

$$s_circ := \begin{bmatrix} 0.7 \\ 0.7 \end{bmatrix}$$

omega-system start [0.5; 0.5].

$$s_om := \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$

3. Local deterministic solvers (bfgs, lbfgs, cg, gd, newton, nm)

optim_bfgs on Rosenbrock -> (1, 1).

$$o_bfgs := \mathrm{optim_bfgs}(f_rosen, s_rn) = \begin{bmatrix} 1 \\ 1 \\ 5.2641 \cdot 10^{-18} \\ 31 \\ 1 \end{bmatrix}$$

optim_lbfgs on Rosenbrock -> (1, 1).

$$o_lbfgs := \mathrm{optim_lbfgs}(f_rosen, s_r0) = \begin{bmatrix} 1 \\ 1 \\ 6.3331 \cdot 10^{-19} \\ 21 \\ 1 \end{bmatrix}$$

optim_cg on Booth -> (1, 3).

$$o_cg := \mathrm{optim_cg}(f_booth, s_b) = \begin{bmatrix} 1 \\ 3 \\ 6.6505 \cdot 10^{-20} \\ 2 \\ 1 \end{bmatrix}$$

Not for commercial use

Created using a free version of SMath Studio

`optim_gd on the quadratic -> (2.25, -4.75).`

$$o_gd := \text{optim_gd}(f_quad, s_r0) = \begin{bmatrix} 2.25 \\ -4.75 \\ 1.6681 \cdot 10^{-17} \\ 94 \\ 1 \end{bmatrix}$$

`optim_newton on the quadratic -> (2.25, -4.75).`

$$o_newton := \text{optim_newton}(f_quad, s_r0) = \begin{bmatrix} 2.25 \\ -4.75 \\ 1.9722 \cdot 10^{-31} \\ 1 \\ 1 \end{bmatrix}$$

`optim_nm on Booth -> (1, 3).`

$$o_nm := \text{optim_nm}(f_booth, s_b) = \begin{bmatrix} 1 \\ 3 \\ 6.329 \cdot 10^{-15} \\ 84 \\ 1 \end{bmatrix}$$

`optim_bfgs on the 4-D sphere -> (0, 0, 0, 0).`

$$o_sph4 := \text{optim_bfgs}\left(f_sph4, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -2.7 \cdot 10^{-14} \\ -2.7 \cdot 10^{-14} \\ -2.6 \cdot 10^{-14} \\ -2.7 \cdot 10^{-14} \\ 2.8704 \cdot 10^{-27} \\ 1 \\ 1 \end{bmatrix}$$

4. Stochastic global solvers on multimodal Rastrigin (explicit bounds required)

Signature: optim_de(expr|fn, vars, x0, p_min, p_max [, max_iter]); population 200;

max_iter = generations. ok = 0 is common - judge by f and x.

`optim_de on Rastrigin -> near (0, 0).`

$$o_de := \text{optim_de}\left(f_rast, \begin{bmatrix} u \\ v \end{bmatrix}, s_ras, lo_r, hi_r, 100\right) = \begin{bmatrix} 1.4455 \cdot 10^{-6} \\ -8.8329 \cdot 10^{-8} \\ 4.1606 \cdot 10^{-10} \\ 101 \\ 1 \end{bmatrix}$$

Not for commercial use

Created using a free version of SMath Studio

`optim_de_prmm on Rastrigin -> (0, 0).`

$$o_de_prmm := \text{optim_de_prmm}\left(f_rast, \begin{bmatrix} u \\ v \end{bmatrix}, s_ras, lo_r, hi_r, 100\right) = \begin{bmatrix} -1.1348 \cdot 10^{-9} \\ 8.5568 \cdot 10^{-10} \\ 0 \\ 1001 \\ 0 \end{bmatrix}$$

`optim_pso on Rastrigin -> (0, 0).`

$$o_pso := \text{optim_pso}\left(f_rast, \begin{bmatrix} u \\ v \end{bmatrix}, s_ras, lo_r, hi_r, 100\right) = \begin{bmatrix} 9.3866 \cdot 10^{-10} \\ -1.1254 \cdot 10^{-9} \\ 0 \\ 100 \\ 0 \end{bmatrix}$$

`optim_pso_dv on Rastrigin -> near (0, 0).`

$$o_pso_dv := \text{optim_pso_dv}\left(f_rast, \begin{bmatrix} u \\ v \end{bmatrix}, s_ras, lo_r, hi_r, 100\right) = \begin{bmatrix} 0.0072 \\ 0.0388 \\ 0.3068 \\ 100 \\ 0 \end{bmatrix}$$

5. Constrained solver (sumt)

The problem is a column [f; c1; ...; cm] with ci(x) <= 0; both a CAS function

and a raw expression form are shown.

`optim_sumt on c_quad -> (5, 4), f = 0.`

$$o_sumt_f := \text{optim_sumt}\left(c_quad, \begin{bmatrix} u \\ v \end{bmatrix}, s_c\right) = \begin{bmatrix} 5 \\ 4 \\ 1.362 \cdot 10^{-25} \\ 10 \\ 1 \end{bmatrix}$$

`optim_sumt on the inline constrained expression -> (5, 4).`

$$o_sumt_e := \text{optim_sumt}\left(\begin{bmatrix} (u-5)^2 + (v-4)^2 \\ (0-2)\cdot u - v + 14 \end{bmatrix}, \begin{bmatrix} u \\ v \end{bmatrix}, s_c\right) = \begin{bmatrix} 5 \\ 4 \\ 1.362 \cdot 10^{-25} \\ 10 \\ 1 \end{bmatrix}$$

Not for commercial use

Created using a free version of SMath Studio

6. Root solvers (broyden, broyden_df) on CAS root systems

F(x) must be a square column [r1; ...; rm], m = n; f holds max|ri|.

`optim_broyden on sys1 -> (0.7071068, 0.7071068).`

$$o_broy1 := \text{optim_broyden}\left(sys1, \begin{bmatrix} u \\ v \end{bmatrix}, s_circ\right) = \begin{bmatrix} 0.7071 \\ 0.7071 \\ 1.1102 \cdot 10^{-16} \\ 13 \\ 1 \end{bmatrix}$$

`optim_broyden_df on sys1 -> same root.`

$$o_broy1df := \text{optim_broyden_df}\left(sys1, \begin{bmatrix} u \\ v \end{bmatrix}, s_circ\right) = \begin{bmatrix} 0.7071 \\ 0.7071 \\ 2.2204 \cdot 10^{-16} \\ 11 \\ 1 \end{bmatrix}$$

`optim_broyden on sys2 (omega constant) -> (0.5671433, 0.5671433).`

$$o_broy2 := \text{optim_broyden}\left(sys2, \begin{bmatrix} u \\ v \end{bmatrix}, s_om\right) = \begin{bmatrix} 0.5671 \\ 0.5671 \\ 2.2204 \cdot 10^{-16} \\ 13 \\ 1 \end{bmatrix}$$

`optim_broyden on the inline residual expression -> (0.7071068, 0.7071068).`

$$o_broy_e := \text{optim_broyden}\left(\begin{bmatrix} u^2 + v^2 - 1 \\ u - v \end{bmatrix}, \begin{bmatrix} u \\ v \end{bmatrix}, s_circ\right) = \begin{bmatrix} 0.7071 \\ 0.7071 \\ 1.1102 \cdot 10^{-16} \\ 13 \\ 1 \end{bmatrix}$$

7. Generic solver, point evaluation, a function name and a single equation

`Generic optim_solve_expr(algo 0 = bfgs, f_rosen, vars, x0) -> (1, 1).`

$$o_solve_e := \text{optim_solve_expr}\left(0, f_rosen, \begin{bmatrix} u \\ v \end{bmatrix}, s_rn\right) = \begin{bmatrix} 1 \\ 1 \\ 5.2641 \cdot 10^{-18} \\ 31 \\ 1 \end{bmatrix}$$

`optim_eval_expr at (2,3): (2-1)^2 + 4*(3-2)^2 = 5.`

$$ev_pt := \text{optim_eval_expr}\left((u-1)^2 + 4 \cdot (v-2)^2, \begin{bmatrix} u \\ v \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = 5$$

Not for commercial use

Created using a free version of SMath Studio

Single-equation start [0.5].

$s_w := [\,0.5\,]$

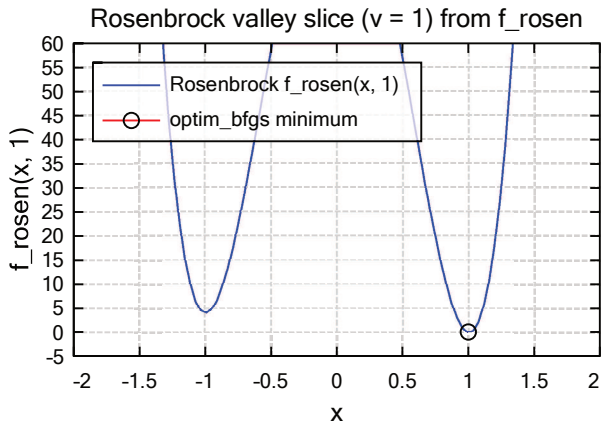
Single transcendental equation w = cos(w) as least-squares -> w = 0.7390851.

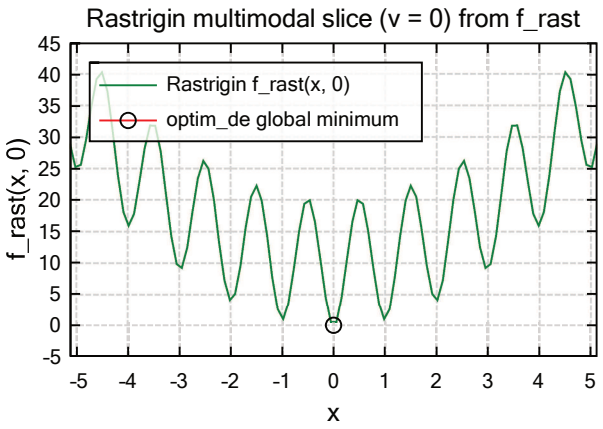
$$o_cosfix := \text{optim_lbfgs}\left(\left(w - \cos(w)\right)^2, [\,w\,], s_w\right) = \begin{bmatrix} 0.7391 \\ 1.245 \cdot 10^{-24} \\ 4 \\ 1 \end{bmatrix}$$

8. Visualisation

CAS benchmark landscapes (with the optimum marked) and solver outcome comparisons.

Series are inline (Chart2D series-inline law).



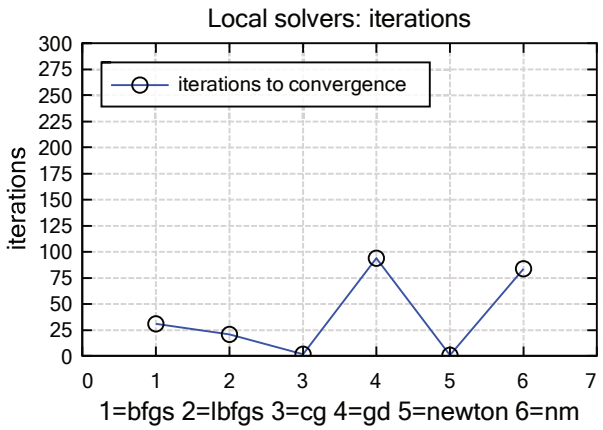


$$\left\{ f_rosen(x, 1) \right\} \left\{ \left[o_bfgs_1 \;\; f_rosen\left(o_bfgs_1, 1\right) \right] \right\}$$

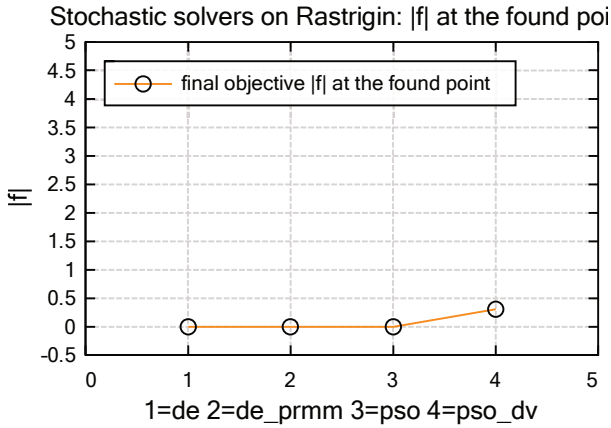
$$\left\{ f_rast(x, 0) \right\} \left\{ \left[o_de_1 \;\; f_rast\left(o_de_1, 0\right) \right] \right\}$$

Not for commercial use

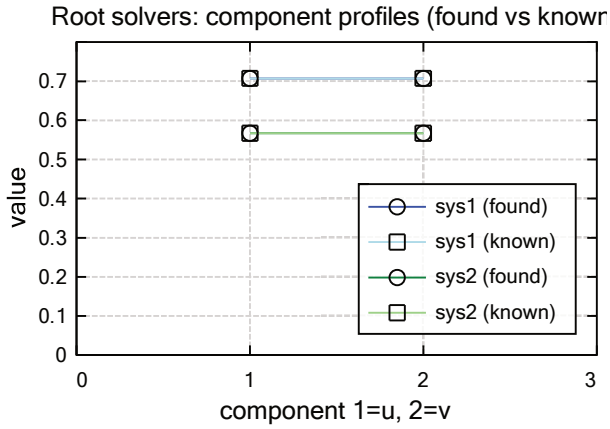
Created using a free version of SMath Studio



$$\text{augment}\left(\begin{bmatrix}1\\2\\3\\4\\5\\6\end{bmatrix},\begin{bmatrix}o_bfgs_4\\o_lbfgs_4\\o_cg_4\\o_gd_4\\o_newton_4\\o_nm_4\end{bmatrix}\right)$$



$$\text{augment}\left(\begin{bmatrix}1\\2\\3\\4\end{bmatrix},\begin{bmatrix}|o_de_3|\\|o_de_prmm_3|\\|o_pso_3|\\|o_pso_dv_3|\end{bmatrix}\right)$$



$$\begin{bmatrix}\text{augment}\left(\begin{bmatrix}1\\2\end{bmatrix},\begin{bmatrix}o_broy1_1\\o_broy1_2\end{bmatrix}\right)\\ \text{augment}\left(\begin{bmatrix}1\\2\end{bmatrix},\begin{bmatrix}0.7071068\\0.7071068\end{bmatrix}\right)\\ \text{augment}\left(\begin{bmatrix}1\\2\end{bmatrix},\begin{bmatrix}o_broy2_1\\o_broy2_2\end{bmatrix}\right)\\ \text{augment}\left(\begin{bmatrix}1\\2\end{bmatrix},\begin{bmatrix}0.5671433\\0.5671433\end{bmatrix}\right)\end{bmatrix}$$

Not for commercial use